# Shrestha Pandey

*Software engineer · AI systems, security, simulation*

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**13** projects shipped since Jul 2026 | **11** live in production | **629** commits | **87,953** lines of code | **1,577** automated tests

## EDUCATION

**VIT Bhopal University** · *B.Tech, Computer Science and Engineering* — *2025 – 2029 · Bhopal*

## EXPERIENCE

**Founder & Lead Engineer** · *Syntrophy* branchhq — *Jul 2026 – Present*

- Building **Branch HQ**, an Electron platform where six specialised AI agents plan, write, review and run software; every change lands as a pull request carrying its own evidence (scan, tests, review, model, approver).
- Wrote **56,700 lines of code over 498 commits**, backed by **1,425 test cases**; **27 of 32** previously unseen build requests run end to end.
- Designed the four-check gate on every build: a deterministic AST security scan, review by a model that never wrote the code, a real launch with a record read back, and a hash-chained audit log; runs on any model, cloud or on-prem.

## SELECTED PROJECTS

**Kakeya Records · new best-known constructions in mathematics** kakeya-records — `Python · NumPy · Numba`

- Solo entry at the 4th Open Scientific Intelligence Hackathon (Oct 2026): directed an AI research agent to **new best-known thin-triangle Kakeya constructions at every published size** (N = 32, 64, 128), beating **Google DeepMind's AlphaEvolve by 6.88%** and GPT-5.4 Pro by 2.05% at N = 128; first record in **63 minutes on an 8 GB laptop**.
- Every record certified twice, by the HorizonMath benchmark's official validator and by an **exact rational-arithmetic verifier**; a prior-art search found no published construction that beats it.

**Zero-Trust Vault** zerotrustvault.in — `TypeScript · WebCrypto · Argon2id/WASM · WebAuthn`

- Zero-knowledge vault: key-committed AES-256-GCM, unlocked by any **K of N Shamir shards (2–16)**, a 256 MiB Argon2id passphrase, both, or a hardware passkey (WebAuthn PRF).
- Identifies forged shards among more than K submitted; isolates keys in a Web Worker under strict CSP and Trusted Types; a daily public CI job proves hosted and offline builds are **byte-identical**.

**MEND · Reed–Solomon, made visible** mend-azure.vercel.app — `C++ → WebAssembly · TypeScript`

- Implemented GF($2^8$) arithmetic and a full RS(255,191) decoder (syndromes, Berlekamp–Massey, Chien, Forney) in C++ compiled to WASM; exact recovery in **10,000 randomized trials** within the error bound.
- Repairs a **5 MB file in under a second** in a Web Worker; WASM runs **1.9–3.3× faster** than a TypeScript port after profiling one hot loop made decoding **8× faster**.

**SPHEREx Moving Sky** · *NASA Space Apps 2026* spherex-moving-sky.vercel.app — `Python · FITS · JPL`

- Finds moving objects in NASA SPHEREx all-sky images by range-reading only the needed rows of 69 MB FITS files; recovered **94% (83/88) of bright known asteroids** against **2,762 JPL predictions**, and tested Planet X detection on dwarf planet Makemake at 52.7 au.

**Bheed Drishti · crowd-safety digital twin** live — `TypeScript · WebGL · social-force sim`

- Simulates thousands of pilgrims in a temple town twice from identical arrivals, A/B-testing a fixed barricade plan against an AI controller that **forecasts density 3 minutes ahead** and acts on gates, one-way lanes and PA, for the 2027 Simhastha Kumbh.

**NCII Shield · privacy-preserving takedowns** live — `Python · FastAPI · BK-tree · Web Workers`

- Victims register a **64-bit perceptual hash, never the image**; a BK-tree matches re-uploads within Hamming distance 10, and an IT Rules 2021 notice is drafted only after the victim confirms.

**Also shipped:** RING0, an OS simulator with six CPU schedulers and paging, deadlock and sync labs (53 tests); Conclave, a CLI where Claude leads Gemini and Codex as independent advisers; Friday, a voice-driven macOS agent with permission-gated tool calling, local voices and speech-to-text, and Gemini or local Ollama brains (110 tests); Drishti, a traffic-signal twin for Indian mixed traffic; a CLPS lunar-site browser on JPL Horizons (NASA Space Apps); Chronos, an in-browser CRDT engine over vector clocks.

## COMPETITIONS

**4th Open Scientific Intelligence Hackathon 2026**, solo: new best-known Kakeya constructions · **NASA Space Apps Challenge 2026**, two entries (SPHEREx, CLPS) · **IEEE SMCS Search & Rescue Competition 2026**, Team Terminal 3: multi-robot rescue system in Python

## SKILLS

**Languages** TypeScript, Python, C++, JavaScript, SQL · **AI systems** multi-agent orchestration, LLM tool calling
**Platforms** Electron, React, Node/Express, FastAPI, WebAssembly, Web Workers · **Graphics** Three.js/WebGL, Canvas
**Security** WebCrypto, AES-GCM, Argon2id, Shamir, WebAuthn, CSP · **Engineering** Vitest, GitHub Actions, Vite, Vercel